



**Accredited with “B+” Grade by NAAC,
Certified Under Section 2(f) & 12B of the UGC Act 1956,
An ISO 9001:2015 Certified Institution
Permanently Affiliated to Thiruvalluvar University, Vellore,
Approved by Govt. of Tamil Nadu & AICTE,
Tiruvannamalai – 606603**

Instruction:

In the Farm Management System (FMS), as a farmer, farm manager, or system administrator, I want to use a centralized Salesforce-based platform to record and manage farm, crop, farmer, and buyer information so that all agricultural data is stored securely and systematically. I want to enter and update farm details, crop production records, and buyer transactions easily through a user-friendly interface with proper validation rules to prevent incorrect entries. I also want the system to generate real-time reports and interactive dashboards so that I can analyze farm productivity, monitor sales, and make informed decisions. Additionally, I want role-based access and secure data storage to ensure that sensitive information is protected while enabling smooth collaboration among users. This system will help reduce manual paperwork, minimize errors, improve efficiency, and support data-driven farm management.

1. INTRODUCTION

Project Overview:

The Farm Management System (FMS) is a Salesforce-based application designed to digitize and streamline agricultural operations. It manages data related to farms, crops, farmers, and buyers while providing real-time insights through reports and dashboards. The system helps farmers and managers monitor agricultural activities efficiently and make informed decisions.

Purpose:

The main purpose of this project is to:

- Digitize agricultural data management
- Improve data accuracy and accessibility
- Automate record handling
- Enable real-time monitoring
- Support data-driven farming decisions
- Enhance productivity and transparency

2. IDEATION PHASE

Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Farmer	Maintain farm and crop records	Data is often incomplete or incorrect	Records are kept manually in registers	Frustrated and confused
PS-2	Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and sheets	Stressed and inefficient
PS-3	Buyer	Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	Uncertain and dissatisfied
PS-4	System Admin	Manage user access and system security	Unauthorized access may occur	Access control is handled manually	Concerned and insecure

Problem Statement for a Farm Management System



Section	Problem Description	Challenges
	Many farmers still rely on traditional methods to manage farm activities such as crop planning, irrigation scheduling, inventory tracking, and financial records. These manual processes often lead to inefficient resource use, poor record keeping, and difficulty in monitoring farm performance.	
Section	Challenges	Solution
 Manual Record Keeping  Inefficient Resource Use  Limited Monitoring  Lack of Data Insights	✓ Manual Record Keeping ✓ Inefficient Resource Use ✓ Limited Monitoring ✓ Lack of Data Insights 	Develop a Farm Management System to digitally manage farm operations and improve efficiency. The system will allow farmers to: ✓ Monitor Crops ✓ Manage Resources ✓ Track Expenses & Profits ✓ Get Weather & Market Info

Goal



Develop a digital platform to help farmers make better agricultural decisions!

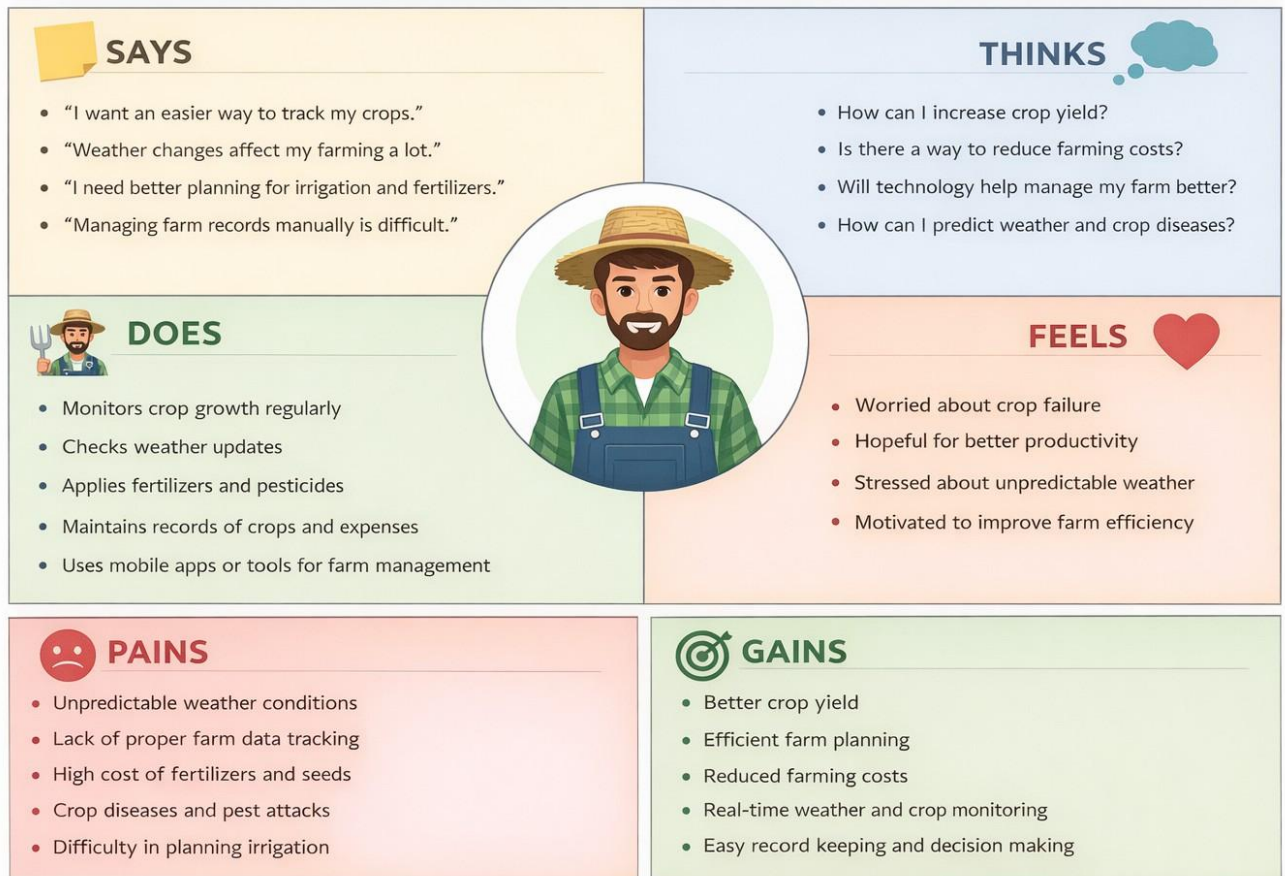
TargetUsers:

- Farmers
- FarmManagers
- Buyers
- SystemAdministrators

EmpathyAnalysis

- **Think&Feel:**Needaccurateandtimelyfarm data
- **Hear:**Feedbackfromagriculturalofficers
- **See:**Registers,paperfiles,spreadsheets
- **Say&Do:**Enterrecordsmanually
- **Pain:**Dataloss,errors,delays
- **Gain:**Automation,fastreporting

EMPATHY MAP – FARM MANAGEMENT SYSTEM



Brainstorming

Brainstormed Ideas

- Manual farm records
- Excel-based management
- Web-based system
- Mobile farm app
- Salesforce-based FMS

Selected Idea

Salesforce-based Farm Management System.

Brainstorming Map for Farm Management System

Possible Features

- ✓ Crop Monitoring
- ✓ Irrigation Scheduling
- ✓ Livestock Tracking
- ✓ Fertilizer Management
- ✓ Weather Updates



Key Problems

- Manual Record Keeping
- Weather Uncertainty
- Crop Monitoring Difficulty
- Market Price Changes
- Resource Wastage



Potential Users

- Farmers
- Farm Owners
- Agricultural Officers
- Farm Workers



Farm Management System

Key Benefits

- Better Productivity
- Cost Control
- Improved Decisions
- Time Saving



Farm Resources

- ✓ Land
- ✓ Seeds
- ✓ Fertilizers
- ✓ Water
- ✓ Equipment
- ✓ Labor



Technology

- Mobile App
- Cloud Database
- GPS Tracking
- IoT Sensors



Key Benefits

- ✓ Better Productivity
- ✓ Cost Control
- ✓ Improved Decisions
- ✓ Time Saving



Brainstorming Goal: Generate ideas and features to create a smart farm management system that enhances efficiency and productivity for farmers.

3.REQUIREMENTANALYSISPHASE

CustomerJourneyMap

Stages:

1. Login
2. AccessFarmManagementApp
3. EnterFarmandCrop Data
4. RegisterFarmersandBuyers
5. GenerateReports
6. AnalyzeResults
7. DecisionMaking

Customer Journey Map for Farm Management System



Stage	Goals	Actions	Feelings	Touchpoints
1.  Identify Needs	Determine what is needed for the upcoming season.	The farmer assesses what to plant, required resources like seeds, fertilizers, water, equipment	Curious, Thoughtful  <i>considering essential needs</i>	- Note-taking - Discussions - Weather Forecast  
2.  Plan & Organize	Plan crops, tasks, and resources efficiently.	The farmer uses the Farm Management System to plan crop planting, schedule irrigation, and allocate resources like labor and equipment.	Organized, Prepared  <i>feeling ready for the season</i>	- Farm Management System dashboard - Planning Tools 
3.  Monitor & Track	Keep track of crop growth, resource usage, and expenses.	The farmer inputs data in the system to monitor crop health, adjust irrigation, track fertilizer use, and record expenses.	Aware, Informed  <i>knowing the farm's status</i>	- Mobile App - Notifications - Weather Updates 
4.  Analyze & Decide	Review data and make informed farming decisions.	The farmer reviews collected data on yields, expenses, and market prices to decide on harvesting and selling crops.	Confident, Strategic  <i>making smart, data-driven decisions</i>	- Reports - Analytics - Market Price Updates 
5.  Evaluate & Improve	Assess results and find areas for improvement.	The farmer evaluates the season's performance and identifies ways to improve crop productivity and reduce costs.	Reflective, Motivated  <i>looking to improve for the future</i>	- Summary Reports - Performance Feedback



Overall Experience: Regular use of the Farm Management System makes farming more organized, efficient, and **profitable**.

SolutionRequirements

FunctionalRequirements

FRNo	Requirement
FR-1	UserAuthentication
FR-2	Farm,Crop,Farmer,BuyerManagement
FR-3	DataEntryandValidation
FR-4	RelationshipMapping
FR-5	ReportGeneration

FR-6	DashboardVisualization
FR-7	UserRoleManagement

Non-FunctionalRequirements

NFR	Description	Example
Usability	Easyinterface	SimpleLightningUI
Security	Dataprotection	Profiles&roles
Reliability	Nodataloss	Cloudbackup

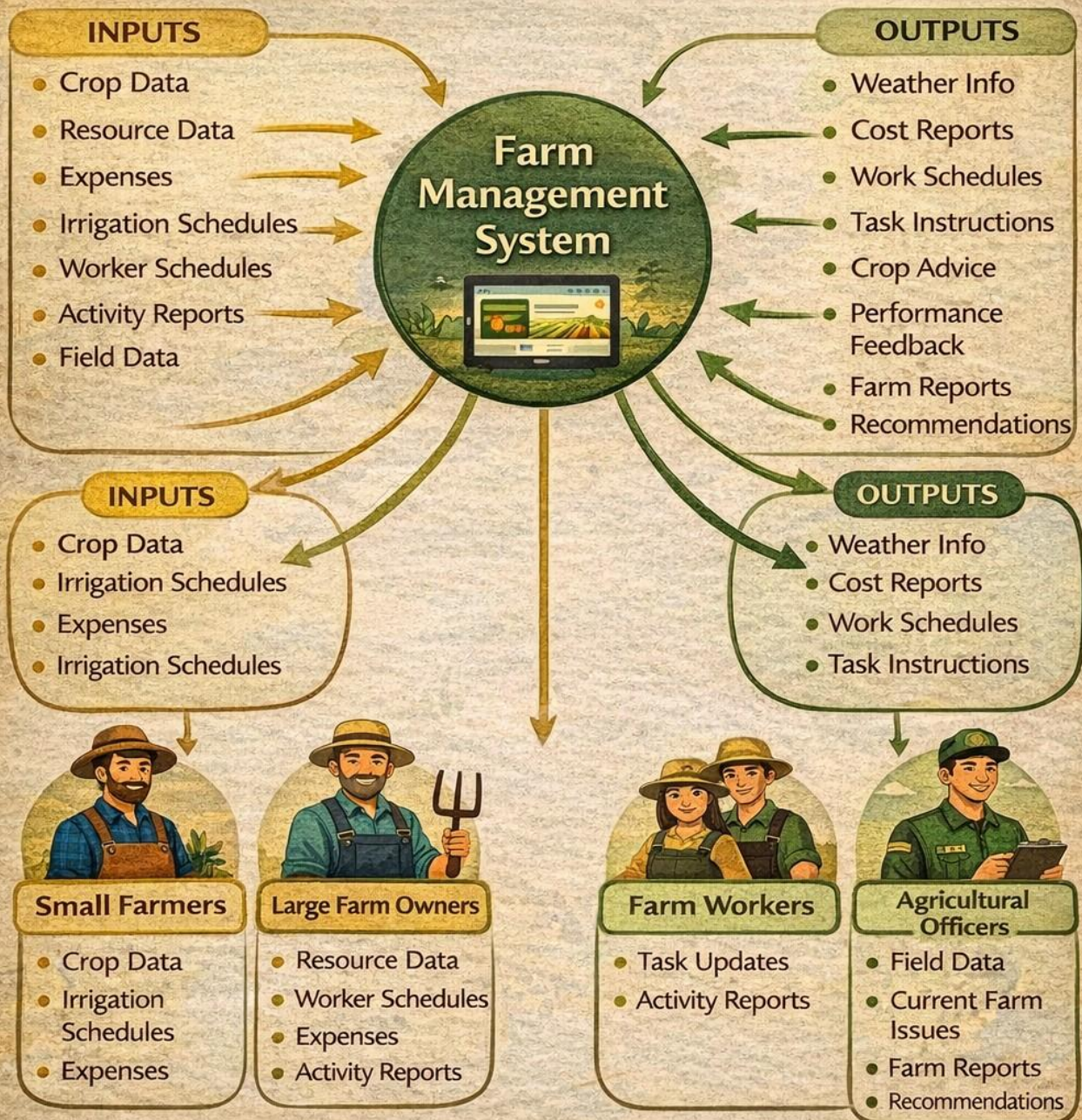
Performance	Fastaccess	Reportsin<3sec
Availability	24/7access	Cloudhosting
Scalability	Handlesgrowth	Morefarms/users

DataFlowDiagram

DataFlow:

User→SalesforceUI→CustomObjects(Farm,Crop,Farmer,Buyer)→Validation Rules →
Relationships → Reports → Dashboards

Data Flow Diagram for Farm Management System



Customers for Farm Management System (FMS) in a context-level Data Flow Diagram; points farm management system to plan crops, manage resources, track farm activities, and improve productivity.

TechnologyStack

Layer	Technology
Frontend	SalesforceLightningUI
Backend	SalesforcePlatform
Database	SalesforceCloud
Automation	ValidationRules
Analytics	Reports&Dashboards

4. PROJECT DESIGN PHASE

Problem–Solution Fit

The FMS replaces manual farming records with automated data management. Cloud storage prevents data loss, dashboards improve analysis, and real-time reports eliminate delays. This ensures accurate and timely agricultural insights.



Proposed Solution

Functional Requirements

Parameter	Description
Problem	Manual maintenance of farm, crop, farmer, and buyer records leading to errors, data loss, and delays.

Solution	Salesforce-based Farm Management System with centralized data storage and automation.
Innovation	Automated data entry validation, cloud storage, and real-time reporting dashboards.
Impact	Improved data accuracy, faster reporting, and better decision-making for farmers and managers.
Business Model	Digital farm management and agricultural data services platform.
Scalability	Supports multiple farms, users, and regions with cloud-based infrastructure.

Solution Architecture

Architecture Layers

1. Presentation Layer – Salesforce Lightning UI
2. Application Layer – Validation Rules, Relationships
3. Data Layer – Custom Objects
4. Analytics Layer – Reports and Dashboards

5. PROJECT PLANNING & SCHEDULING

Project Planning

- Created Salesforce Developer Org
- Developed custom objects
- Configured tabs and Lightning App
- Created fields and relationships
- Designed profiles and roles
- Created users and sharing rules
- Developed reports and dashboards
- Conducted testing and reviews

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	UserStory/ Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure a Salesforce developeraccountsothatIcansetup and deploy the Farm Management System.	3	High	Jayamoorthy v
Sprint-2	DataModeling	USN-2	As an admin, I want to create Farm, Crop, Farmer, and Buyer objects with properrelationshipssothatagricultural data is stored and managedefficiently.	5	High	Vimal E
Sprint-2	DataModeling	USN-3	As a staff member, I want tabs and a LightningAppforallfarmmodulesso that I can easily navigate and perform daily operations.	5	High	Santhosh R
Sprint-3	Automation	USN-4	Asasystem,Iwanttovalidatefarm and crop details such as dates, quantities, and mandatory fields so that incorrect or incompleterecords areprevented.	3	High	Monesh K
Sprint-3	Automation	USN-5	As a manager, I want automated flows for crop updates and buyer notificationssothatfarmoperations areprocessedfasterwithoutmanual intervention.	3	High	Harish E
Sprint-4	Security	USN-6	As an administrator, I wantrole-based access control for farmers,managers,andbuyersso that sensitive farm and transaction data is secure.	5	High	JayamoorthyV
Sprint-5	Reports	USN-7	As a farm manager, I want reports showingcropyield,sales,andbuyer transactions so that I can monitor farm performance and makeinformeddecisions.	4	High	Vimal E
Sprint-6	Dashboards	USN-8	As a farm manager, I want dashboards showingcropproductivity,salestrends, and operational statistics so that I can monitor performance visually.	4	Medium	Santhosh R

ProjectTracker,Velocity&BurndownChart:(4Marks)

Sprint	TotalStory Points	Duration	SprintStartDate	SprintEndDate (Planned)	StoryPoints Completed(as on Planned End Date)	SprintRelease Date (Actual)
Sprint 1	20	6Days	07 Feb2026	13 Feb2026	20	13 Feb2026
Sprint 2	20	6Days	15 Feb2026	21 Feb2026	18	20 Feb2026
Sprint 3	20	6Days	02 Mar2026	08 Mar2026	19	05 Mar2026
Sprint 4	20	6Days	10 Mar2026	16 Mar2026	16	16 Mar2026

6. ProjectDevelopmentPhase:

ProjectDescription:

TheFarmManagementSystem(FMS)inSalesforceisacomprehensivesolution designedtodigitizeandstreamlineagriculturaloperations.Thisapplicationempowers farmers, farmmanagers, and stakeholders by managing data related to crops, Farm, Farmer, and buyers, while providing actionable insights through reports, and dashboards.

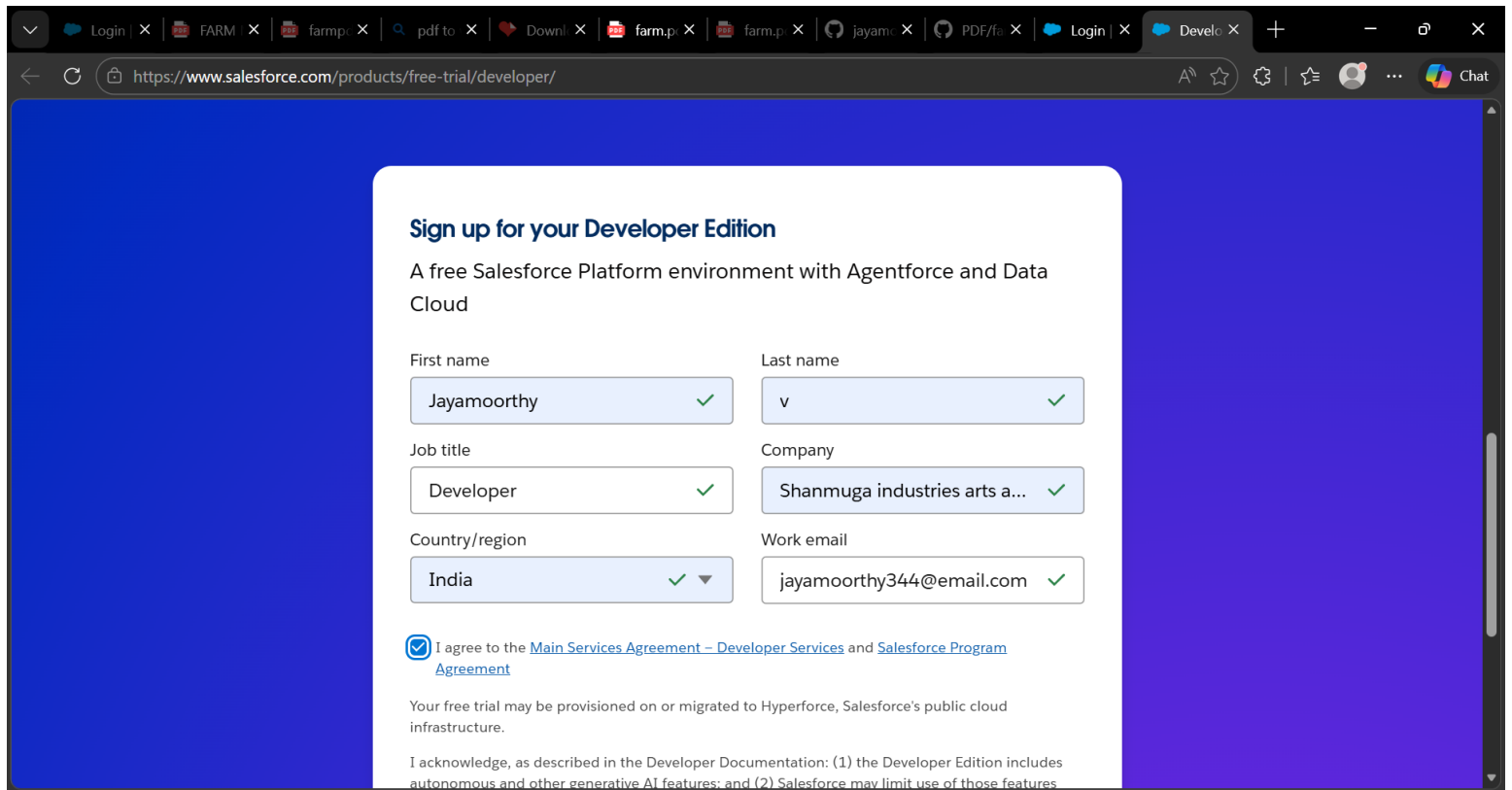
Milestone1-Salesforce:

WhatIsSalesforce?

Salesforceisyourcustomersuccessplatform,designedtohelpyousell,service,market, analyze, and connect with your customers.

Salesforcehaseverythingyouneedtorunyourbusinessfromanywhere.Usingstandard products and features, you can manage relationships with prospects and customers, collaborateandengagewithemployeesandpartners,andstoreyourdatasecurelyinthe cloud.

Singup Developer Account



The screenshot shows the Salesforce Developer Edition sign-up page. The browser's address bar displays the URL <https://www.salesforce.com/products/free-trial/developer/>. The page has a blue background with a white sign-up form in the center. The form includes fields for First name (Jayamoorthy), Last name (v), Job title (Developer), Company (Shanmuga industries arts a...), Country/region (India), and Work email (jayamoorthy344@email.com). Each field has a green checkmark indicating it is valid. Below the form, there is a checkbox for agreeing to the Main Services Agreement, Developer Services, and Salesforce Program Agreement, which is checked. A note states that the free trial may be provisioned on or migrated to Hyperforce, Salesforce's public cloud infrastructure. At the bottom, a disclaimer acknowledges the Developer Edition features and potential limitations on generative AI features.

Sign up for your Developer Edition

A free Salesforce Platform environment with Agentforce and Data Cloud

First name: Jayamoorthy ✓

Last name: v ✓

Job title: Developer ✓

Company: Shanmuga industries arts a... ✓

Country/region: India ✓

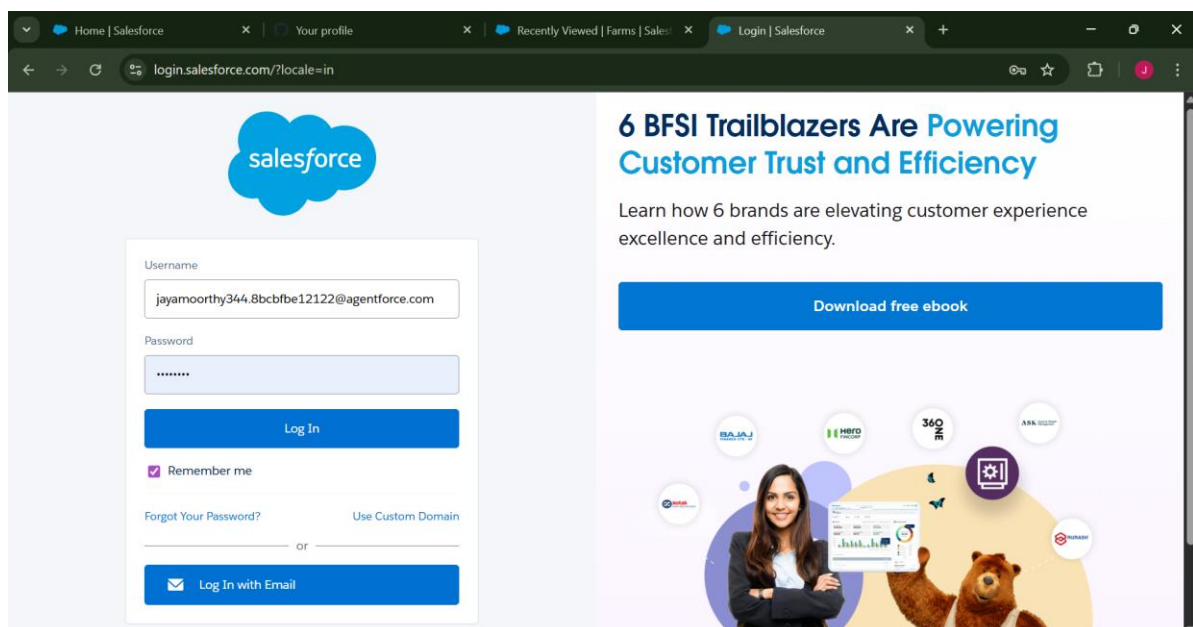
Work email: jayamoorthy344@email.com ✓

☒ I agree to the [Main Services Agreement – Developer Services](#) and [Salesforce Program Agreement](#)

Your free trial may be provisioned on or migrated to Hyperforce, Salesforce's public cloud infrastructure.

I acknowledge, as described in the Developer Documentation: (1) the Developer Edition includes autonomous and other generative AI features; and (2) Salesforce may limit use of those features

Login Developer Account



The screenshot shows the Salesforce login page. The browser's address bar displays the URL login.salesforce.com/?locale=in. The page features the Salesforce logo at the top left. On the left side, there is a login form with fields for Username (jayamoorthy344.8bcbf12122@agentforce.com) and Password (masked with dots). Below the password field is a blue 'Log In' button. There is a 'Remember me' checkbox which is checked, and links for 'Forgot Your Password?' and 'Use Custom Domain'. At the bottom of the form is a 'Log In with Email' button. On the right side, there is a promotional banner for '6 BFSI Trailblazers Are Powering Customer Trust and Efficiency' with a 'Download free ebook' button. The banner includes a woman holding a tablet and several circular icons representing different BFSI brands.

salesforce

Username: jayamoorthy344.8bcbf12122@agentforce.com

Password: [masked]

Log In

☒ Remember me

[Forgot Your Password?](#) [Use Custom Domain](#)

Log In with Email

6 BFSI Trailblazers Are Powering Customer Trust and Efficiency

Learn how 6 brands are elevating customer experience excellence and efficiency.

Download free ebook

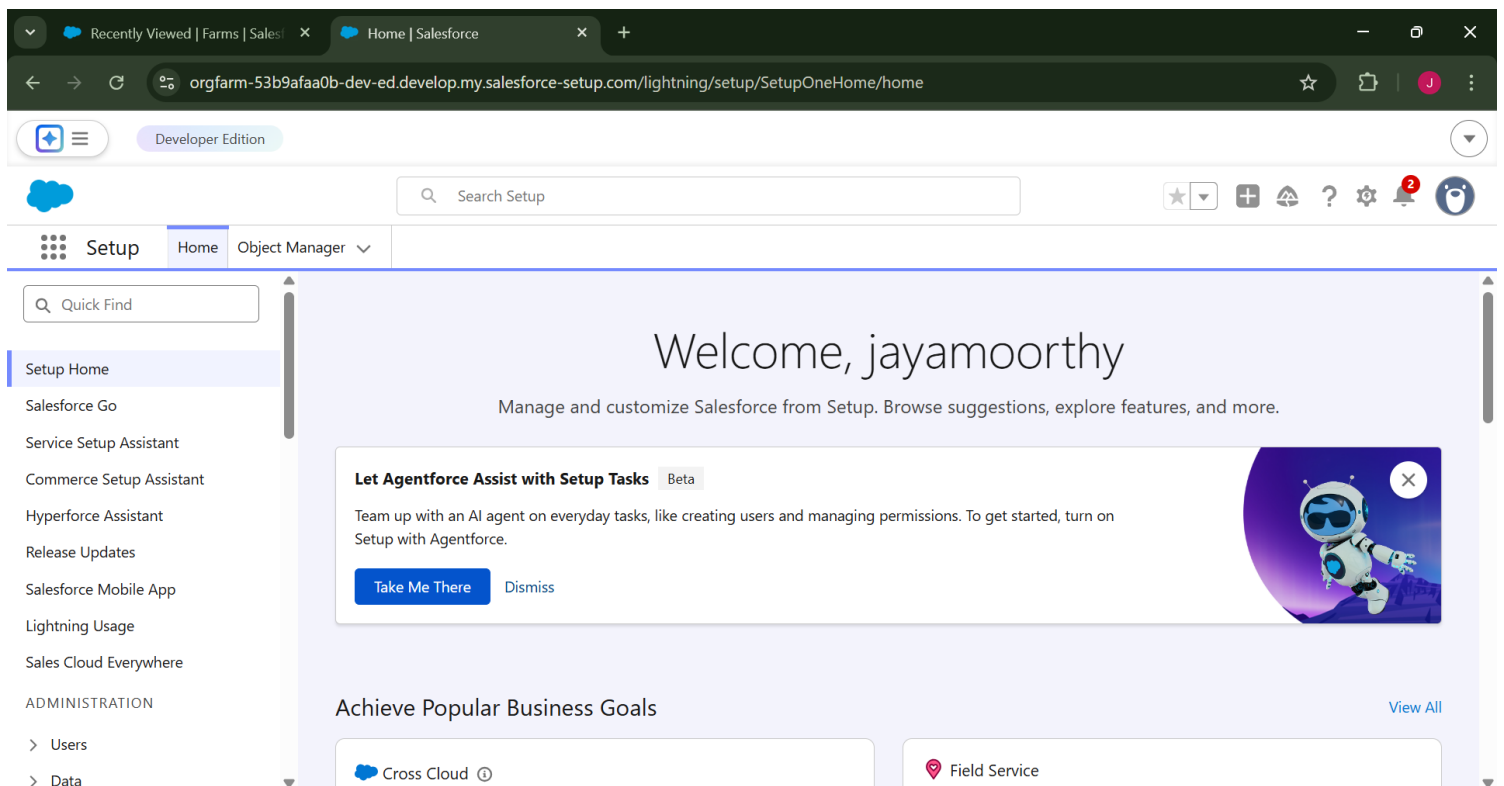
Activity1:

Creating a Salesforce Developer Org:

A Developer org has all the features and licenses you need to get started with Salesforce.

1. Search [Developer.salesforce.com/signup](https://developer.salesforce.com/signup)
2. Enter the following details like Firstname, lastname, Email, Role, Company, Country/Region, Postal code, and Username must be unique.
3. Click sign me up, after a few mins you will receive an email from Salesforce org and by using the verify account link you can create your new password.
4. Click save.
5. Search login.salesforce.com

By using username and password you can log into the Salesforce org. The homepage will appear below



Milestone2-Object:

What is an object?

Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

In This Application We Use 4 Custom Objects:

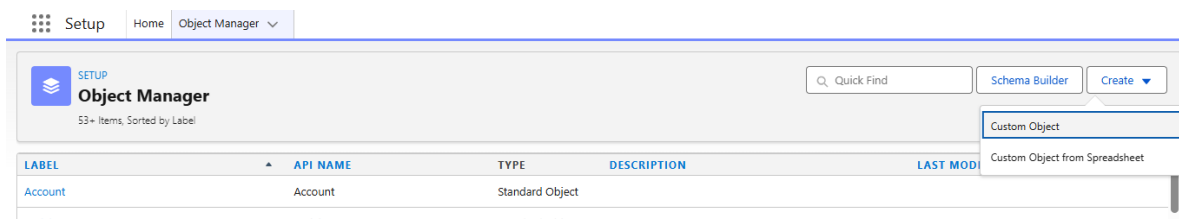
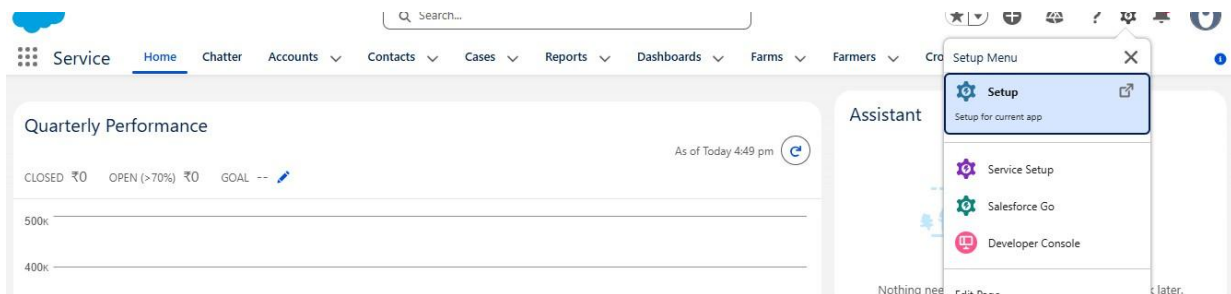
1. Farm
2. Crop
3. Farmer
4. Buyer

Activity1:

Create a custom object for Farm:

To create a custom object, follow these steps:

1. From setup click on object manager.
2. Click create, select custom object.



3. Fill in the label as "Farm".
4. Fill in the plural label as "Farms".
5. Record name: FarmName
6. Select the data type as "Text".

SETUP

New Custom Object

Optional Features

☒ Allow Reports

☒ Allow Activities

☒ Track Field History

☐ Allow in Chatter Groups

☐ Enable Licensing [i](#)

Object Classification

When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object. [Learn more](#).

☒ Allow Sharing

☒ Allow Bulk API Access

☒ Allow Streaming API Access

Deployment Status

☐ In Development

☒ Deployed

[What is this?](#)

Search Status

7. In the Optional Features section, select Allow Reports and Track Field History.
8. In the Deployment Status section, ensure Deployed is selected.
9. In the Search Status section, select Allow Search.
10. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
11. Leave everything else as is, and click Save.

Activity 2:

Creation of Crop object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Crop
 - c. Plural Label: Crops
 - d. Record Name: CropName
 - e. Select the data type as "Text".
 - f. Check the Allow Reports checkbox. Check the Allow Search checkbox
 - h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout
 - i. Click Save.

Activity 3:

Creation of Farmer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Farmer

- c. PluralLabel:Farmers
- d. RecordName:FarmerName e. Select the data type as "Text".
- f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.
- h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout. i. Click Save.

Activity4:

Creation of Buyer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Buyer
 - c. Plural Label: Buyers
 - d. Record Name: Buyer Name
 - e. Select the data type as "Text".
 - f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.
 - h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
 - i. Click Save.

Milestone3-

Tab: Tab

Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

Activity1:

Creation of Farm tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.

Developer Edition

Search Setup

★

▼

+

🔍

?

⚙️

🔔

Setup

Home

Object Manager ▼

o

Feature Settings

Analytics

Tableau

Tableau Embedding

Tableau UAF Claims Definition

User Interface

Console Settings

Loaded Console Tab Limit

Rename Tabs and Labels

Custom Object Tabs

SETUP

Tabs

Custom Tabs

Help for this Page

You can create new custom tabs to extend Salesforce functionality or to build new application functionality.

Custom Object tabs look and behave like the standard tabs provided with Salesforce. Web tabs allow you to embed external web applications and content within the Salesforce window. Visualforce tabs allow you to embed Visualforce pages. Lightning Component tabs allow you to add Lightning components to the navigation menu in Lightning Experience and the mobile app. Lightning Page tabs allow you to add Lightning Pages to Lightning Experience and the mobile app.

Custom Object Tabs

New What Is This?

Action	Label	Tab Style	Description
Edit Del	Buyers	People	
Edit Del	Crops	Apple	

2. Under custom object tabs, click New.

Setup

Home

Object Manager ▼

tab

Feature Settings

Analytics

Tableau

Tableau Embedding

Tableau UAF Claims Definition

User Interface

Console Settings

Loaded Console Tab Limit

Rename Tabs and Labels

Custom Object Tabs

SETUP

Tabs

Edit Custom Object Tab

Farms

Help for this Page

Fill in the fields below to define the custom tab.

Custom Tab Definition Edit

Custom Object Tab Information

Required Information

Tab Label	Farms
Object	Farm
Tab Style	Apple

(Optional) Choose a Home Page Custom Link to show as a splash page the first time your users click on this tab.

Splash Page Custom Link

--None--

Enter a short description.

Description

In't find what you're looking for? Try

3. For Object, select Farm.

For Tab Style, select any icon.

4. Leave all defaults as is. Click Next, Next, and Save.

Activity 2:

Creation of Crop tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.

2. Under custom object tabs, click New.

3. For Object, select Crop.

4. For Tab Style, select any icon.

5. Leave all defaults as is. Click Next, Next, and Save.

Activity 3:

Creation of Farmer tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.

2. Under custom object tabs, click New.

3. For Object, select Farmer.

4. For Tab Style, select any icon.

5. Leave all defaults as is. Click Next, Next, and Save.

Activity 4:

Creation of Buyer tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.

2. Under custom object tabs, click New.

3. For Object, select Buyer

4. For Tab Style, select any icon.

5. Leave all defaults as is. Click Next, Next, and Save.

Milestone 4-Lightning App:

What is an App?

Apps in Salesforce are a group of tabs that help the application function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Activity 1:

Create the Farm Management System app

The screenshot shows the Salesforce Lightning Experience App Manager interface. The left sidebar contains navigation links: Setup, Home, and Object Manager. Below these is a search bar with 'app' entered. The sidebar also lists various app categories: Salesforce Mobile App, Data (Mass Transfer Approval Requests), Apps (App Manager, AppExchange Marketplace), Connected Apps (Connected Apps OAuth Usage, Manage Connected Apps), and External Client Apps (External Client App Manager). The main content area is titled 'Lightning Experience App Manager' and displays a table of 34 items. The table is sorted by App Name and filtered by All app menu items. The table columns are: App Name, Developer Name, Description, Last Modified, App Type, and Visibility. The table lists 9 items, including All Tabs, Analytics Studio, Ant Migration Tool, App Launcher, Approvals, Automation, Bolt Solutions, Community, and Content.

	App Name ↑	Developer Name	Description	Last Modified ...	App Type	Vi...
1	All Tabs	AllTabSet		03/03/2026, 12:16 pm	Classic	
2	Analytics Studio	Insights	Build CRM Analytics dashboards and apps	03/03/2026, 12:16 pm	Classic	✓
3	Ant Migration To...	Forcecom_Migration_Tool	The Force.com Migration Tool is a Java/Ant-based co...	03/03/2026, 12:16 pm	Connected (Managed)	
4	App Launcher	AppLauncher	App Launcher tabs	03/03/2026, 12:16 pm	Classic	✓
5	Approvals	Approvals	Manage approvals and approval flows	03/03/2026, 12:16 pm	Lightning	✓
6	Automation	FlowsApp	Automate business processes and repetitive tasks.	03/03/2026, 12:26 pm	Lightning	✓
7	Bolt Solutions	LightningBolt	Discover and manage business solutions designed fo...	03/03/2026, 12:16 pm	Lightning	✓
8	Community	Community	Salesforce CRM Communities	03/03/2026, 12:16 pm	Classic	✓
9	Content	Content	Salesforce CRM Content	03/03/2026, 12:16 pm	Classic	✓

- From Setup, enter App Manager in the Quick Find and select App Manager.
 - Click New Lightning App. Farm Management System as the App Name, then click Next
 - Under App Options, leave the default selections and click Next.
 - Under Utility Items, leave as is and click Next.
 - From Available Items, select Farm, Crop, Farmer, Buyer Reports, and Dashboards and move them to Selected Items. Click Next.
 - From Available Profiles, select System Administrator and move it to Selected Profiles. Click Save & Finish.
 - To verify your changes, click the App Launcher, type Farm Management System and select the Job Application app.
- Note:
1. App Launcher-Displays available apps.
 2. App Name-Displays the current selected app.

Developer Edition

New Lightning App

App Details & Branding

Give your Lightning app a name and description. Upload an image and choose the highlight color for its navigation bar.

App Details

*App Name ⓘ
Name your app...

*Developer Name ⓘ
Enter a developer name...

Description ⓘ
Enter a description...

App Branding

Image ⓘ
Upload

Primary Color Hex
Value ⓘ
#0070D2

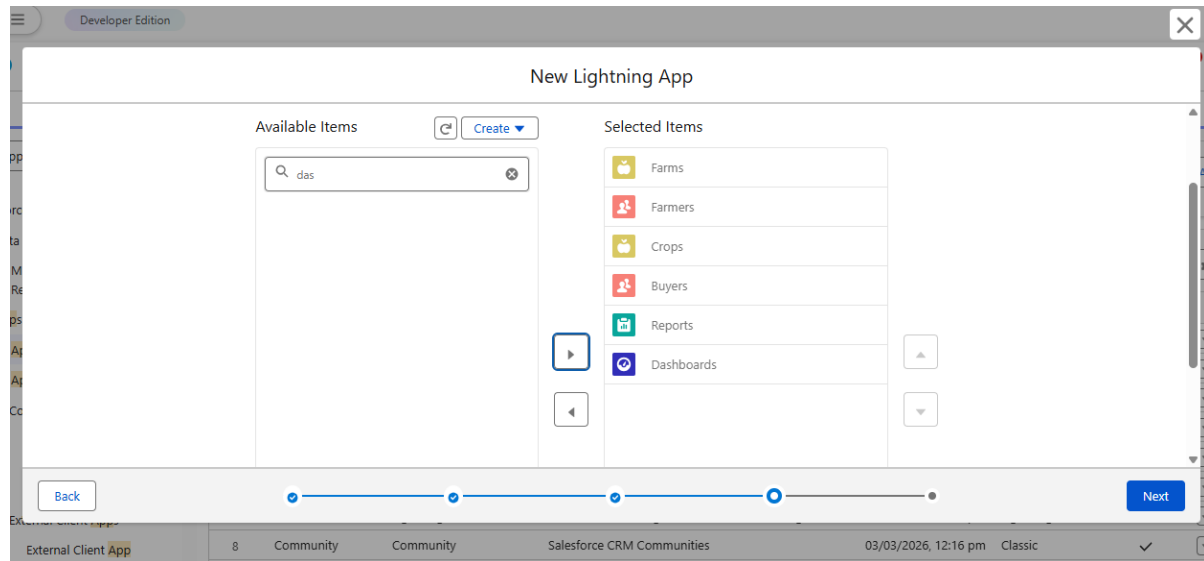
Oro Theme Options

Next

External Client App Manager

8 Community Community Salesforce CRM Communities 03/03/2026, 12:16 pm Classic

3. Navigationmenu-Displaysthetabsavailableinsidetheapp.



Milestone5- FieldsandRelationship:Whatarefields?

FieldsinSalesforcerepresentwhatthecolumnsrepresentinrelationaldatabases.Itcan store data values which are required for a particular object in a record.

Activity1:

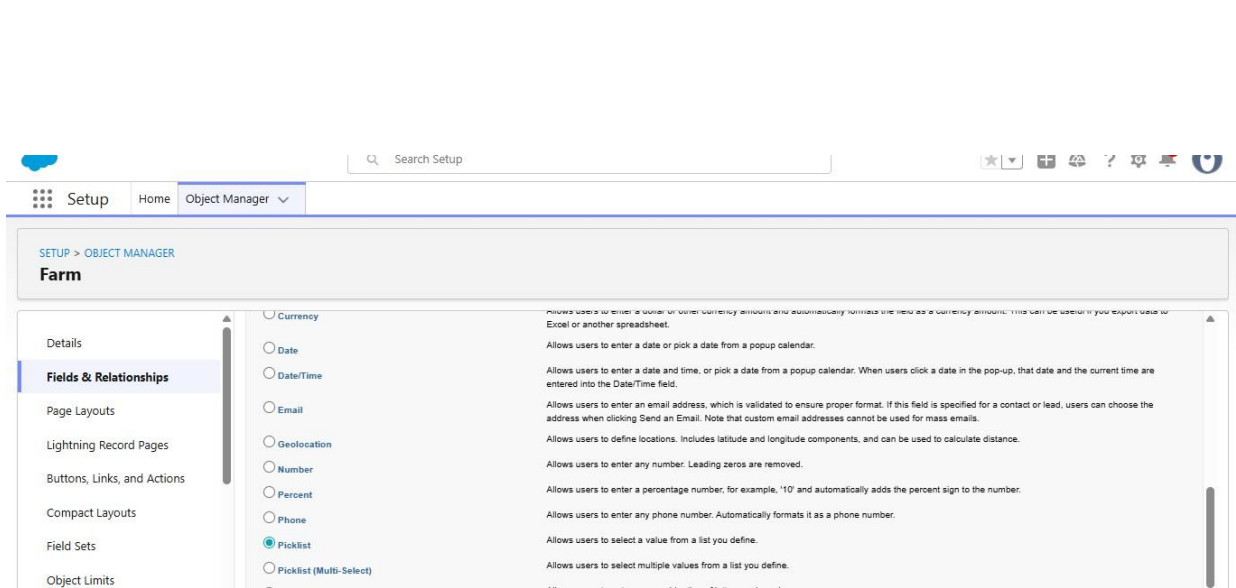
CreationoffieldsfortheFarmobjects:

1. clickthegeariconandselectSetup.ThislaunchesSetupin anew tab.
2. ClicktheObjectManagertabnexttoHome.
3. SelectFarm.
4. SelectFields&Relationshipsfromtheleftnavigation,andclickNew

The screenshot shows the Salesforce Setup interface. The top navigation bar includes the Setup icon, a search bar, and various utility icons. The left sidebar contains navigation links for Setup, Home, and Object Manager. The main content area is titled 'Farm' and displays the 'Fields & Relationships' section. This section includes a table with columns for Field Label, Field Name, Data Type, Controlling Field, and Indexed. The table lists several fields for the Farm object, including 'Created By', 'Farm Type', 'Farmer', 'Form Name', and 'Irrigation type'.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Farm Type	Farm_Type__c	Picklist		
Farmer	Farmer__c	Master-Detail(Farmer)		✓
Form Name	Name	Text(80)		✓
Irrigation type	Irrigation_type__c	Picklist		

5. Choose Datatype as a picklist, Click next



6. EnterFieldLabel(FarmType),andEnterValuesas(Crop,Livestock,Mixed)
- 7.Next,Next Click Save.

ICT MANAGER

Farm New Custom Field [Help](#)

Step 2. Enter the details [Previous](#)

Field Label [i](#)

Values ☐ Use global picklist value set

☒ Enter values, with each value separated by a new line

crop

Livestock

Mixed

Now let's create the other fields following Activity 1 steps 1 to 4 and we must choose the data types of the fields carefully.

1. Next we create another picklist field, enter field label (Soil Quality) enter values (Organic, Cover Crops, Composting)
2. Again we create picklist field, field label (Irrigation type), enter values (Sprinkler irrigation, Drip irrigation, localized irrigation)
3. Choose data type as text, enter field label (Location)
4. Choose data type as text, enter field label (size)
5. Create a master-detail relationship with Farmer Object.

Activity 2:

Creation of fields for the Crop objects:

1. Choose data type as text, enter field label name (Crop Name), next save.
2. Select data type as date, enter field label (Planting Date), then save.
3. Again we select data type as date, enter field label (Harvest date), save.
4. Next we create a lookup relationship on crop object related to Farm Object.
5. We select data type as currency, enter field label (Actual Yield), next save it.
6. Select data type as a number, enter field label (Total Acres Harvested), enter maximum length (20) and save.
7. Now we create a formula field for that select data type as a formula, click next.

Setup

Home

Object Manager

SETUP > OBJECT MANAGER

Crop

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Data Type

☐ None Selected

Select one of the data types below.

☐ Auto Number

A system-generated sequence number that uses a display format you define. The number is automatically incremented for each new record.

☒ Formula

A read-only field that derives its value from a formula expression you define. The formula field is updated when any of the source fields change.

☐ Roll-Up Summary

A read-only field that displays the sum, minimum, or maximum value of a field in a related list or the record count of all records listed in a related list.

☐ Lookup Relationship

Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list.

☐ Master-Detail Relationship

Creates a special type of parent-child relationship between this object (the child, or "detail") and another object (the parent, or "master") where:

- The relationship field is required on all detail records.
- The ownership and sharing of a detail record are determined by the master record.
- When a user deletes the master record, all detail records are deleted.
- You can create rollup summary fields on the master record to summarize the detail records.

The relationship field allows users to click on a lookup icon to select a value from a popup list. The master object is the source of the values in the list.

☐ External Lookup Relationship

Creates a relationship that links this object to an external object whose data is stored outside the Salesforce org.

8. EnterFieldLabel(ExpectedYieldperAcre),Selectformulareturtypeas Currency,click Next.
9. EnterFormulaActual,thenclicknextand save.

Enter your formula and click Check Syntax to check for errors. Click the Advanced Formula subtab to use additional fields, operators, and functions.

Example: [More Examples...](#)

Select Field Type
Insert Field

ExpectedYieldperAcre (Currency) =

Actual_Yield__c

Activity3:

CreationoffieldsfortheFarmerobjects:

1. SelectDatatypePhone,Enterfieldlabel(ContactNumber),nextSave.
2. SelectDatatypePicklist,EnterFieldLabel(State),Entervalue(Telangana,Andhra Pradesh,Tamilnadu),save.
3. ChooseDatatypePicklist,EnterFieldLabel(District),EnterValuesofdistrict,nextsave.
4. AgainselectDatatypeasPicklist,EnterFieldLabel(Village),EnterValuesof villages,next save.

Creationoffielddependency:

1. NowcreateFieldDependencyForDistrictfield,OpenObjectManager,ChooseFarmer object ,select Fields&Relationships,On Right side We have Field Dependency option,select that option.
2. Clicknew,SelectControllingFieldasState,DependentFieldAsDistrict,Clickon Continue.
3. SelecttheappropriateValueineachcolumnbydouble-clickingthem.
4. ForEx.Telangana-YadadriBhongir.
5. ClickIncludeValues.AnditisalsothesameforAndhraPradesh,Tamilnaduwithits city.
6. ClickPreview,thentestthependencybyselectingadifferentStateandviewingthe associate Values available for Particular state.



Search Setup



Setup

Home

Object Manager

SETUP > OBJECT MANAGER

Farmer

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

New Field Dependency

[Help for this Page](#)

Create a dependent relationship that causes the values in a picklist or multi-select picklist to be dynamically filtered based on the value selected by the user in another field.

- The field that drives filtering is called the "controlling field." Standard and custom checkboxes and picklists with at least one and less than 300 values can be controlling fields.
- The field that has its values filtered is called the "dependent field." Custom picklists and multi-select picklists can be dependent fields.

Step 1. Select a controlling field and a dependent field. Click Continue when finished.

Step 2. On the following page, edit the filter rules that control the values that appear in the dependent field for each value in the controlling field.

Continue Cancel

Controlling Field State

Dependent Field --None--

Continue Cancel

Developer Edition

Search Setup

Setup Home Object Manager

Quick Find

Setup Home

Salesforce Go

Service Setup Assistant

Commerce Setup Assistant

Hyperforce Assistant

Release Updates

Salesforce Mobile App

Lightning Usage

Sales Cloud Everywhere

ADMINISTRATION

> Users

> Data

Showing Columns: 1 - 3 (of 3) < Previous | Next > View All

State:	Telangana	Andhra Pradesh	Tamilnadu
District:	Hyderabad	Hyderabad	Hyderabad
	Secunderabad	Secunderabad	Secunderabad
	Bhongir	Bhongir	Bhongir
	Nalgonda	Nalgonda	Nalgonda
	Vishakapatnam	Vishakapatnam	Vishakapatnam
	Vijayawada	Vijayawada	Vijayawada
	Nellore	Nellore	Nellore
	Amaravati	Amaravati	Amaravati
	Chennai	Chennai	Chennai
	Coimbatore	Coimbatore	Coimbatore
	Kanchipuram	Kanchipuram	Kanchipuram
	Kanniyakumari	Kanniyakumari	Kanniyakumari

Showing Columns: 1 - 3 (of 3) < Previous | Next > View All

Click button to include or exclude selected values from the dependent picklist:

Include Values Exclude Values

7. Click Close to close the preview window, Click Save.

Activity4:

Creation of fields for the Buyer objects:

1. Select Datatype as a Phone, Enter a Field Label (Contact), Save.
2. Select Datatype as an Email, Enter Field Label (Email), Next save.
3. Select Datatype Textarea, Enter Field Label (Address), Save it.
4. Select Datatype Picklist, Field Label (Buyer Type), Enter values (Wholesale, Retail), Click Save.

Milestone6-Profile:

What is a profile?

A profile is a group/collection of settings and permissions that define what a user can do in

salesforce. A profile controls “Object permissions, Field permissions, User permissions, Tab settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types, Login hours & Login IP ranges. A profile can be assigned to many users, but a user can be assigned a single profile at a time.

Activity1:

Create a custom profile:

1. From setup, enter profiles in Quick Find box
2. Select profiles (Standard user).
3. Click clone.
4. For Profile, enter Sales Department.

Developer Edition

Search Setup

Setup

Home

Object Manager

Q pro

Hyperforce Assistant

Users

Profiles

Data

Mass Transfer Approval Requests

Feature Settings

Approval Settings

Data.com

Prospector Preferences

Prospector Users

Functions

Marketing

Lead Processes

SETUP

Profiles

Profiles

Help for this Page

All Profiles

Edit

Delete

Create New View

New Profile

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Other All

Action	Profile Name	User License	Custom
☐ Edit Clone	Customer Portal Manager Custom	Customer Portal Manager Custom	☐
☐ Edit Clone	Customer Portal Manager Standard	Customer Portal Manager Standard	☐
☐ Edit Clone	External Apps Login User	External Apps Login	☐
☐ Edit Clone	External Identity User	External Identity	☐
☐ Edit Del I...	Farm Manager	Salesforce	☒
☐ Edit Clone	Force.com - App Subscription User	Force.com - App Subscription	☐
☐ Edit Clone	Force.com - Free User	Force.com - Free	☐
☐ Edit Clone	Gold Partner User	Gold Partner	☐

1-25 of 43

0 Selected

Previous Next

Page 1 of 2

Farm Management System

FarmsFarmersCropsBuyersReportsDashboards

Farmers

Recently Viewed

NewImportChange OwnerAssign Label

3 items • Updated a few seconds ago

Search this list...

Farm Name

1

Gayathri

2

ShivaShanker

3

Shiva Shanker

Farm Management System

FarmsFarmersCropsBuyersReportsDashboards

Buyer

Logesh

New ContactEditNew Opportunity

RelatedDetails

Buyer Name

Logesh

Contact

(887) 766-5443

Email

Address

Buyer Type

Wholesale

Created By

jayamoorthy V, 3/6/2026, 9:56 PM

Owner

jayamoorthy V

Last Modified By

jayamoorthy V, 3/6/2026, 9:56 PM

Activity

Filters: All time • All activities • All types

RefreshExpand AllView All

Upcoming & Overdue

No activities to show.

Get started by sending an email, scheduling a task, and more.

To change what's shown, try changing your filters.

Show All Activities

Activity2:

Create a custom profile-2:

Create a profile with the profile name as “FarmManager”.

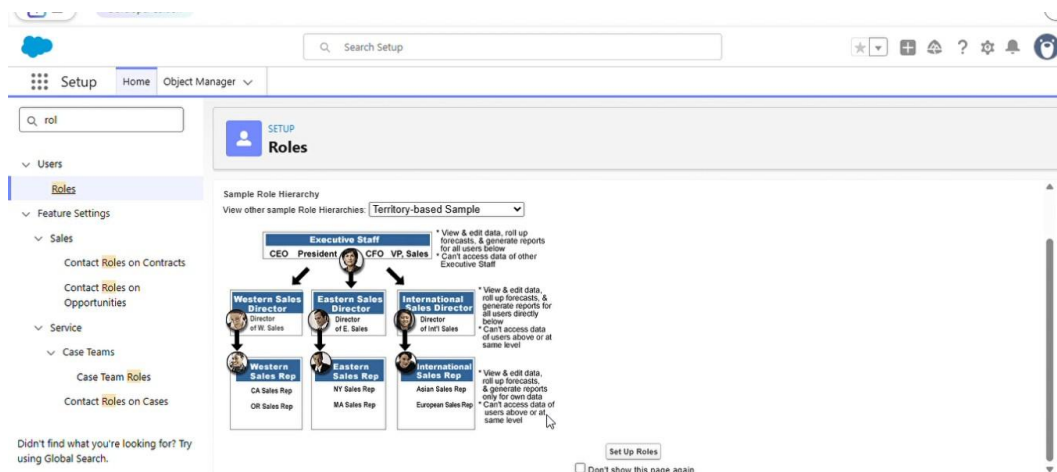
5. From setup, enter profiles in QuickFind box
6. Select profiles (Standard user).
7. Click clone.

Milestone7-Role

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

Activity-1 Creation of role

1. From the QuickFind box search for the role and click on the roles option
2. select the setup roles option



3. Below the CEO click on add role and enter the label name as a "Farm Manager" and role name will be Automatically populated and click on save.

The screenshot displays the Salesforce Setup interface for the 'Roles' section. The left sidebar contains a navigation menu with 'Roles' highlighted. The main content area is titled 'Creating the Role Hierarchy' and includes instructions on building the hierarchy. Below this, a tree view shows the existing hierarchy: 'Shanmuga industries arts and science college' is the root, with a sub-role 'CEO' listed below it. The 'CEO' role has an 'Add Role' link next to it. The interface also includes a search bar at the top, a 'Show in tree view' dropdown, and a footer message: 'Didn't find what you're looking for? Try using Global Search.'

Milestone8-User:

What is a user?

A user is anyone who logs into Salesforce. Users are employees at your company, such as sales reps, managers, and IT specialists, who need access to the company's records. Every user in Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.

NOTE- As a Salesforce license can only be used by 2 users at a time in Dev Org, so if you don't find a Salesforce license, then deactivate a user who has a Salesforce license or change the license type from Salesforce to any other.

Activity1:

To Create a user:

1. From Setup, enter Users in the Quick Find box, then select Users.
2. Click New User.

Search Setup

Setup Home Object Manager

users

Users

Analytics Groups

Permission Set Groups

Permission Sets

Profiles

Public Groups

Queues

Roles

User Management Settings

Users

Feature Settings

Data.com

Prospector Users

Didn't find what you're looking for? Try using Global Search.

SETUP Users

User Profile Help for this Page

User: santhosh monesh

Permission Set Assignments (0) | Permission Set Assignments: Activation Required (0) | Permission Set Group Assignments (0) | Permission Set License Assignments (0) | Personal Groups (0) | Public Group Membership (0) | Queue Membership (0) | Team (0) | Managers in the Role Hierarchy (0) | OAuth Apps (0) | Third-Party Account Links (0) | Built-in Authenticators (0) | Installed Mobile Apps (0) | Authentication Settings for External Systems (0) | Login History (0+) | User Provisioning Accounts (0)

User Detail

Edit Sharing Reset Password Freeze View Summary

Name	santhosh monesh	Role	Sales Department
Alias	smone	User License	Salesforce
Email	sksanthovai.143@gmail.com [Verify] [i]	Profile	Sales Department
Username	moneshsanthosh.143@gmail.com	Active	☒
Nickname	User17728535165712174392 [i]	Marketing User	☐
Title		Offline User	☐
Company		Knowledge User	☐
Department		Flow User	☐
Division		Service Cloud User	☐
Address		Site.com Contributor User	☐
Time Zone	(GMT-08:00) Pacific Standard Time (America/Los_Angeles)	Site.com Publisher User	☐
Locale	English (United States)	WDC User	☐
Language	English	Mobile Push Notifications	View

3. Enter Firstname as Far and lastname as Manager.
4. Enter the user's name and email address and a unique username in the form of an email address. By default, the username is the same as the email address.
5. Select the role Farm Manager.
6. Select user License as Standard Platform User.
7. Select profile (Farm Manager).
8. Click save.

Activity 2:

Create a user with a username as "Sales Department", and assign him the sales Manager profile. Follow the steps from above Activity

Milestone 9-Sharing Rules:

What are Sharing Rules?

Sharing rules help users to share records based on conditions. It is basically created for objects whose organization-wide defaults (OWD) are set to public read-only or private because sharing rules can only extend the access and not restrict it.

Types of sharing rules,

- Owner-based Sharing Rules • Criteria-based Sharing Rules

Activity 1:

Create a sharing rule

- 1) Go to Sharing Settings, which can be found under the Quick Find section.
- 2) Scroll down and find the candidate object where a sharing rule needs to be added, and then click on New to create a new sharing rule.
- 3) Add the label of the sharing rule you want to make.
- 4) Select your rule type based on the criteria.
- 5) Select the field can join immediately check field from the Crop object.
- 6) Select the Crop as equal and value is True.
- 7) And in selecting the users to share with this section select roles and in that select Farm Manager.
- 8) And in this section of select the level of access for the users give the access Read/Write.
- 9) And save the rule.

Buyer

Logesh

New Contact

Edit

New Opportunity

Related

Details

Buyer Name

Logesh

Contact

(887) 766-5443

Email

Address

Buyer Type

Wholesale

Created By

jayamoorthy V

, 3/6/2026, 9:56 PM

Owner

jayamoorthy V

Last Modified By

jayamoorthy V

, 3/6/2026, 9:56 PM

Activity

📅

📧

📞

📧

Filters: All time • All activities • All types

Refresh • Expand All • View All

Upcoming & Overdue

No activities to show.
Get started by sending an email, scheduling a task, and more.

To change what's shown, try changing your filters.

Show All Activities

Activity2:

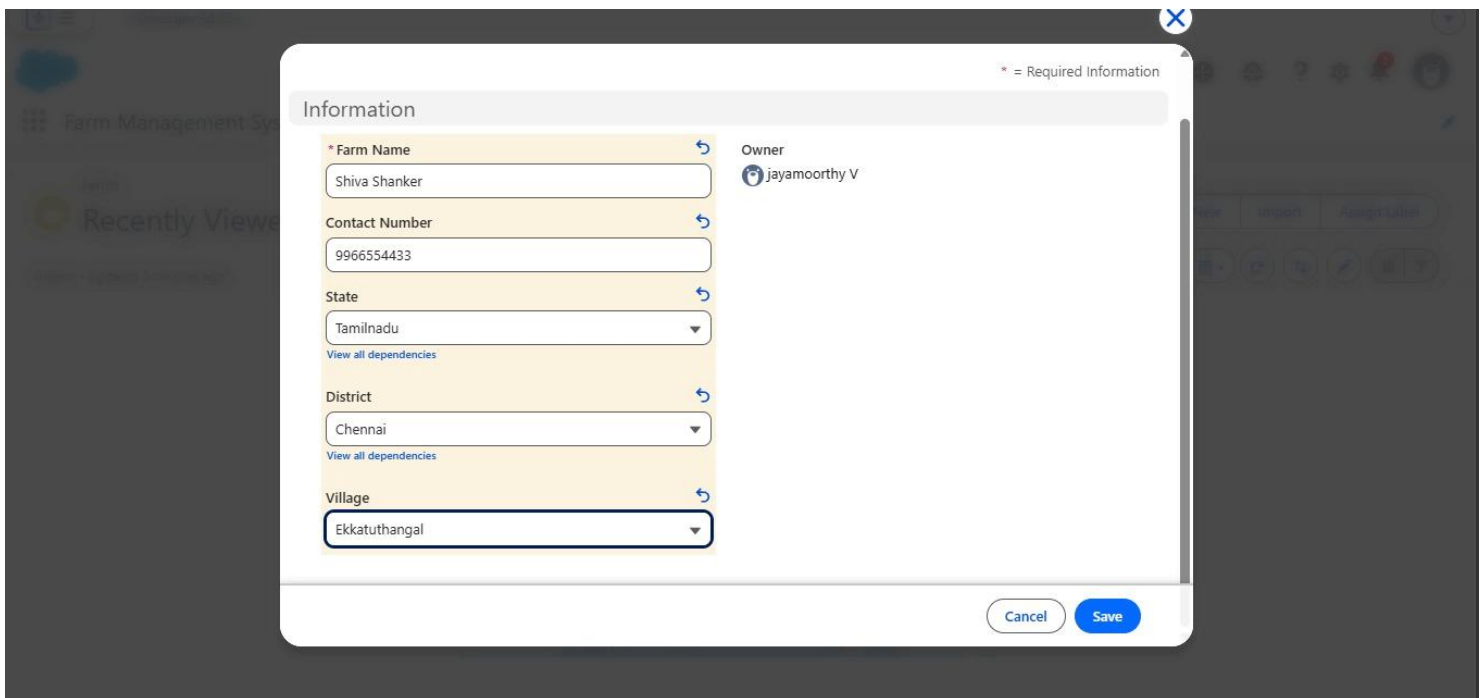
Create a Sharing Rule to Share the records of BuyertoSales department with the Access of Read/Write.

Follow the steps from above Activity.

Milestone-10: User Adoption

Activity1: Create a Record (Farmer)

- Click on App Launcher on left side of screen.
- Search Farm Management & click on it.
- Click on Farmer Tab.
- Click new and fill details & Save.



The screenshot shows a mobile application interface for 'Farm Management System'. A modal form titled 'Information' is displayed, used for creating a new Farmer record. The form includes fields for 'Farm Name' (Shiva Shanker), 'Contact Number' (9966554433), 'State' (Tamilnadu), 'District' (Chennai), and 'Village' (Ekkatuthangal). Each field has a refresh icon. To the right, the 'Owner' is listed as 'jayamoorthy V' with a user icon. A legend at the top right indicates that an asterisk (*) denotes required information. At the bottom right of the form are 'Cancel' and 'Save' buttons. The background shows a blurred view of the app's main interface with a sidebar menu and a 'Recently Viewed' section.

* = Required Information

Information

* Farm Name 

Shiva Shanker

Contact Number 

9966554433

State 

Tamilnadu 

[View all dependencies](#)

District 

Chennai 

[View all dependencies](#)

Village 

Ekkatuthangal 

Owner

 jayamoorthy V

Cancel Save

Milestone 11-Reports:

What are Reports?

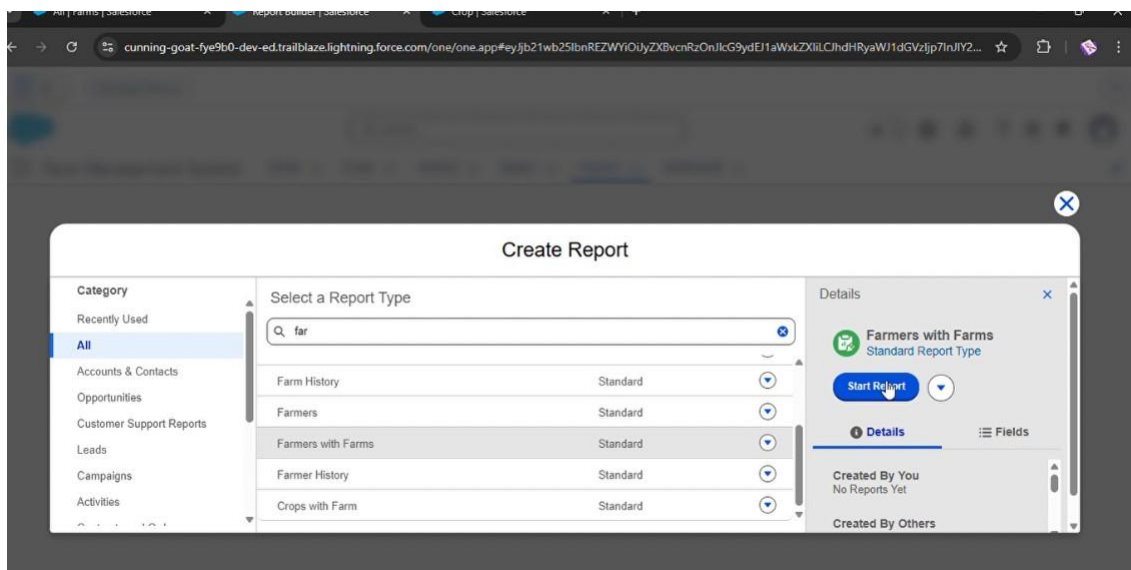
A report is a list of records that meet the criteria you define. It's displayed in rows and columns, and can be filtered, grouped, or displayed in a graphical chart. Every report is stored in a folder. Folders can be public, hidden, or shared, and can be set to read-only or read/write.

Activity 1:

Create report:

Create a report that displays rating of the account and which has type and account name.

- 1) Click on app launcher search for reports.
- 2) Click on the new report and select the category has Farmer with Farms.



- 3) In the details section select the option start report.
- 4) In Outline Click on Column Select Which Field you want to display in the Report.
- 5) Click Group, Select Which Fields You Want to Group.
- 6) Save the report by giving label name (Farmer with Farms) and save the folder as a public folder and save the report.



Q Search...



Farm Management SystemFarmsFarmersCropsBuyersReportsDashboards

ReportsRecent2 items

Q Search recent reports...New ReportNew Folder

REPORTS	Report Name	Description	Folder	Created By	Created On	Subscribed
Recent	Crops with Farm Report		Private Reports	jayamoorthy V	3/6/2026, 10:11 PM	
Created by Me	Farmers with Farms Report		Private Reports	jayamoorthy V	3/6/2026, 10:05 PM	
Private Reports						
Public Reports						
All Reports						
FOLDERS						
All Folders						
Created by Me						
Shared with Me						
FAVORITES						
All Favorites						

Activity2:

Create a Report using the Objects Crops, Buyers. Follow the steps from above Activity.

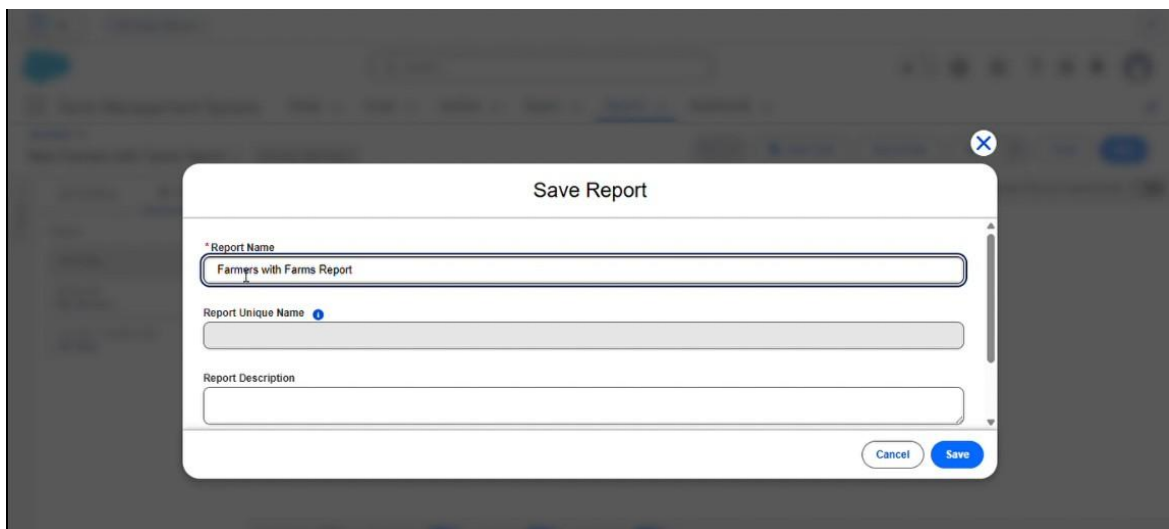
Milestone12-Dashboard:

Dashboards provide more insights than reports as they combine the data from many reports and show a summarized result. Looking at many reports at a time gives the flexibility of combining the results from them quickly. Also summaries in dashboards help us decide on action plans quicker. The dashboards can contain charts, graphs and Tabular data.

Activity1:

Create a Dashboard:

1. Click the Dashboard tab.
2. Click New Dashboard.
3. Name the dashboard Farmers with farms and click Create.
4. Click + Component.
5. Select the Farmers with Farms and click Select.
6. Select the Gauge chart and click Add.
7. Click Save and then Done.



The image shows a 'Save Report' modal window. The title bar says 'Save Report' and has a close button (X). The form inside has three fields: 'Report Name' (containing 'Farmers with Farms Report'), 'Report Unique Name' (empty), and 'Report Description' (empty). At the bottom right are 'Cancel' and 'Save' buttons.

Milestone-13: View Report and Dashboard:

Activity 1: Report

- Click on App Launcher on left side of screen.
- Search Farm Management System & click on it.
- Click on Reports Tab.
- Click on Farmers with Farms & see records

The screenshot displays the 'Farm Management System' dashboard. At the top, there is a search bar and a notification that says 'Report "Farmers with Farms Report" was saved'. Below the search bar, the 'Farmers with Farms Report' is shown. The report includes a table with columns for 'Farm Type', 'Farmer: Farmer Name', 'Farm: Farm Name', 'size', and 'Soil Quality'. The table lists two records: 'Crop (1)' for 'Deepadharshini' and 'Mixed (1)' for 'ShivaShanker'. The 'Total (2)' row shows the sum of the two records.

Report: Farmers with Farms
Farmers with Farms Report

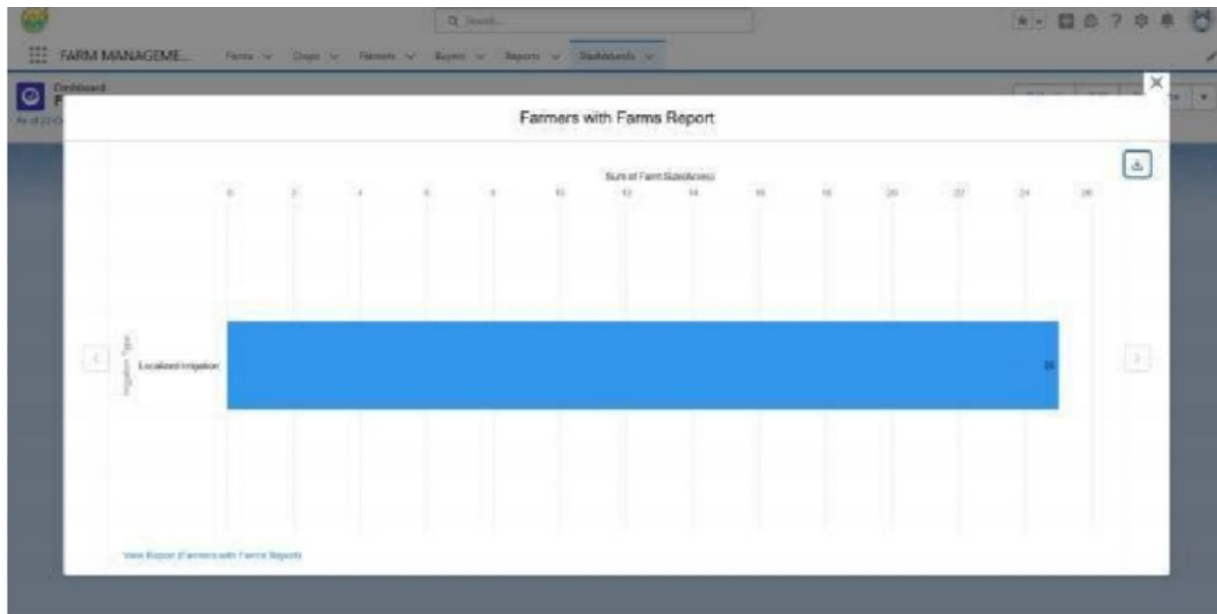
Enable Field Editing Add Chart Edit

Total Records
1

☐ Farm Type ↑	Farmer: Farmer Name	Farm: Farm Name	size	Soil Quality
☐ Crop (1)	Deepadharshini	Deepadharshini Farm	15	Composting
Subtotal				
☐ Mixed (1)	ShivaShanker	ShivaShanker Farm	10	Organic
Subtotal				
Total (2)				

Activity2:Dashboard

- ClickonAppLauncheronleftsideofscreen.
- SearchFarmManagementsystem&clickon it.
- ClickonDashboardTab.
- ClickFarmersWithfarms&seerecords.





Developer Edition



Search...



Farm Management System

Farms

Crops

Farmers

Buyers

Reports

Dashboards



Dashboard

Farm Management dashBoard

As of 03-Mar-2026, 4:58 pm · Viewing as Vennila SV

Crops with Farm Report



[View Report \(Crops with Farm Report\)](#)

As of 03-Mar-2026, 4:58 pm

Farmers with Farms Report



Farm Type

Crop

Mixed

[View Report \(Farmers with Farms R...](#)

As of 03-Mar-2026, 4:58 pm

7. FUNCTIONAL&PERFORMANCETESTING

Duringthetestingphase,screenshotswerecapturedfromtheSalesforceinterfacetoverifysystem functionality.

The following were tested:

- CreationofFarm,Crop,Farmer,andBuyerrecords
- Validationofinputfields
- Relationshipanddependencychecks
- Profileandroleaccesstesting
- Sharingruleverification
- Reportanddashboardgeneration

Performancetestingensuredsmoothsystemresponseandstabilityduringmultipleuser operations.

8. ADVANTAGES&DISADVANTAGES

Advantages

- Centralizeddatasystem
- Reducedpaperwork
- Improvedaccuracy
- Bettermonitoring
- Real-timeanalytics

Disadvantages

- Requiresinternetconnection
- Salesforcelicensedependency
- Limitedofflineaccess

9. Conclusion:

The Farm Management System successfully digitizes agricultural data management using Salesforce. It improves efficiency, reduces errors, and enables farmers and managers to make informed decisions through automation and analytics.

10. FUTURESCOPE

- Mobileapplicationintegration
- AI-basedcropprediction
- Weatherdataintegration
- IoTsensorconnectivity
- Governmentschemeintegration
- Marketpriceforecasting

11. APPENDIX

- SourceCode:SalesforceConfigurationandMetadata
- Dataset:Farm,Crop,Farmer,andBuyerRecords
- ToolsUsed:SalesforceDeveloperOrg

12 .URL

<https://www.salesforce.com/trailblazer/pte88xerrjwt54katb>

Thank You